

Introduction to RMarkdown

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Introduction

- R Markdown provides an unified authoring framework for data science, combining your code, its results, and your prose commentary.
- R Markdown documents are fully reproducible and support dozens of output formats, like PDFs, Word files, slideshows, and more.

Usage of RMarkdown files

1. For communicating to decision makers, who want to focus on the conclusions, not the code behind the analysis.
2. For collaborating with other data scientists (including future you!), who are interested in both your conclusions, and how you reached them (i.e. the code).
3. As an environment in which to do data science, as a modern day lab notebook where you can capture not only what you did, but also what you were thinking.

Prerequisite

prerequisite

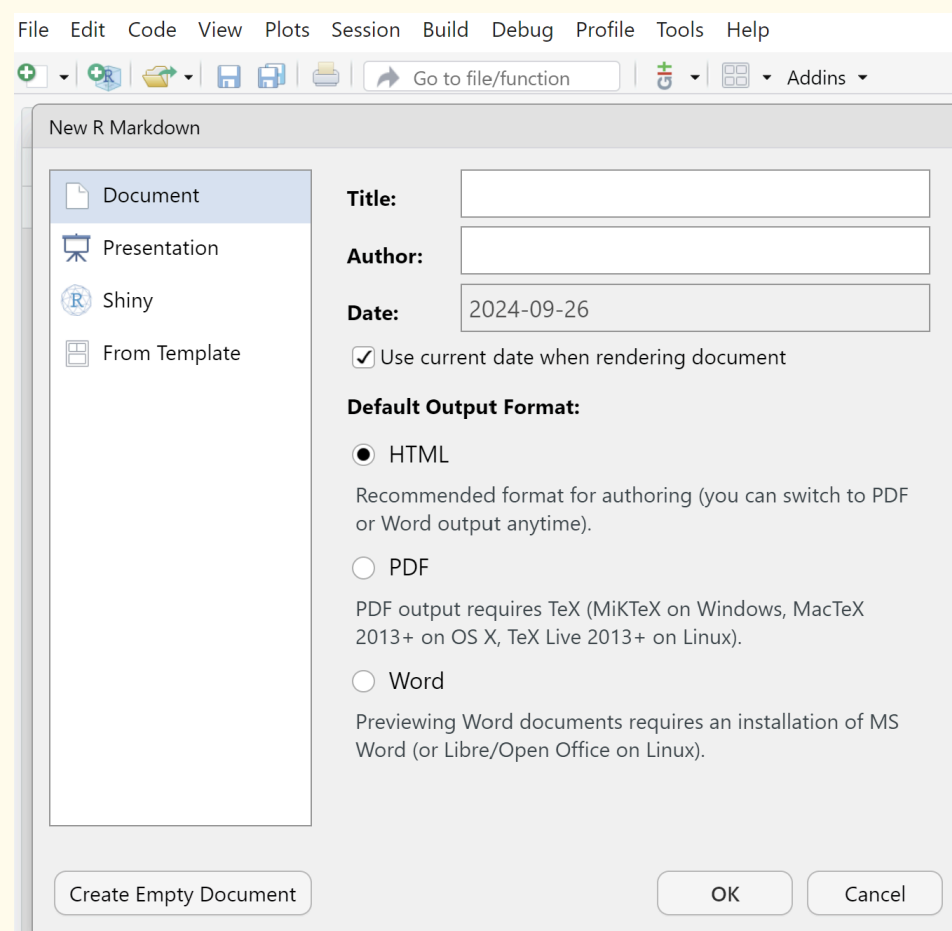
You need the rmarkdown package, but you don't need to explicitly install it or load it, as RStudio automatically does both when needed.

R Markdown Basics

- RMarkdown file is a plain text file that has the extension `.Rmd`
- The basic types of content of RMarkdown file is given as:
 - i. An optional **YAML header** surrounded by `---`
 - ii. **Chunks** of R codes surrounded by `````
 - iii. Text mixed with simple text formatting

HTML Documents

- Markdown was originally designed for HTML output, so it may not be surprising that the HTML format has the richest features among all output formats.
- To create an HTML Document click on File then New file and choose R Markdown



Appearance of the YAML header

- The YAML appears as follows:

```
1 ---
2 title: "Title"
3 author: "Author Details"
4 date: "September 22,2024"
5 output:html_document:
6 ---
```

Table of contents

- Use the `toc` option to include an automatically generated table of contents in the output document.
- Use the `toc-depth` option to specify the number of section levels to include in the table of contents.
- The default is 3 meaning headings 1,2,3 will be included.

```
1 ---
2 title: "Title"
3 author: "Author Details"
4 date: "September 22,2024"
5 output:
6   html_document:
7     toc: true
8     toc_depth: 3
9 ---
```


Floating TOC

- You can specify the `toc_float` option to float the table of contents to the left of the main document content
- The floating table of contents will always be visible even when the document is scrolled.

```
1 ---
2 title: "Title"
3 author: "Author Details"
4 date: "September 22,2024"
5 output:
6   html_document:
7     toc: true
8     toc_float: true
9 ---
```

Section Numbering

- You can add section numbering to headers using the `number_sections` options

```
1 ---
2 title: "Title"
3 author: "Author Details"
4 date: "September 22,2024"
5 output:
6   html_document:
7     toc: true
8     number_sections: true
9 ---
```

Appearance and Style

- `theme` specifies the Bootstrap theme to use for the page (themes are drawn from the Bootswatch theme library). Valid themes include default, bootstrap, cerulean, cosmo, darkly, flatly, journal, lumen, paper, readable, sandstone, simplex, spacelab, united, and yeti.
- `highlights` specifies the syntax highlighting style. Supported styles include default, tango, pygments, kate, monochrome, espresso, zenburn, haddock, breezedark, and textmate. Pass null to prevent syntax highlighting.
- `smart` indicates whether to produce typographically correct output, converting straight quotes to curly quotes, `---` to em-dashes, `--` to en-dashes, and `...` to ellipses. Note that smart is enabled by default.

Example

```
1 ---
2 title: "SPS 2444 Statistical Computing"
3 output:
4   html_document:
5     theme: united
6     highlight: tango
7 ---
```

Custom css

- You can add your own CSS to an HTML document using the `css` option

```
1 ---
2 title: "SPS 2444 Statistical Computing and Reporting"
3 output:
4   html_document:
5     css: styles.css
6 ---
```

- If you want to provide all of the styles for the document from your own CSS you set the theme (and potentially highlight) to null

```
1 ---
2 title: "SPS 2444"
3 output:
4   html_document:
5     theme: null
6     highlight: null
7     css: styles.css
8 ---
```

Figure Options

- There are a number of options that affect the output of figures within HTML documents
 - i. `fig_width` and `fig_height` can be used to control the default figure width and height
 - ii. `fig_retina` specifies the scaling to perform for retina displays
 - iii. `fig_caption` controls whether figures are rendered with captions
 - iv. `dev` controls the graphic device used to render the figure

Example

```
1 ---
2 title: "SPS 2444"
3 output:
4   html_document:
5     fig_width: 7
6     fig_height: 6
7     fig_caption: true
8 ---
```

Data Frame printing

- The following options are available
 - i. `default` printing using the generic method
 - ii. `kable` from the `knitr` package
 - iii. `tibble` use the `tibble::print.tbl_df` function
 - iv. `paged` use `rmarkdown::paged_table` to create a pageable table

References

- You can consult the following reference for more details
1. Xie, Yihui, JJ Allaire, and Jeffrey Horner. 2023. Markdown: Render Markdown with Commonmark. <https://github.com/rstudio/markdown>.